

FOYERS, United Kingdom

WMO: 030570

Lat: 57.267N

Lon: 4.483W

Elev: 69

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	26.6	29.6	20.7	15.5	31.0	23.3	17.6	33.0	28.4	42.9	25.1	44.9	1.0	190	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.7	72.9	61.4	68.9	59.9	65.9	58.4	63.0	69.4	61.3	66.9	59.6	64.6	9.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	78.9	65.7	58.7	73.9	63.7	57.1	69.7	62.4	28.5	69.9	27.3	67.2	26.2	64.7	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	21.2	18.7	23.4	78.3	4.2	2.9	20.4	80.4	17.9	82.1	15.5	83.7	12.4	85.8
			22.7	66.3	3.9	3.0	19.9	68.5	17.6	70.3	15.4	72.0	12.6	74.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	40.0	40.3	42.4	45.5	50.1	54.2	58.3	59.0	54.0	48.3	43.8	39.5	
	DBStd	8.28	4.81	4.92	4.57	4.53	5.19	4.18	3.67	4.31	4.11	4.82	5.27	5.63	
	HDD50	1657	311	273	236	146	61	11	0	0	9	88	193	329	
	HDD65	6219	775	692	700	584	462	325	210	195	331	517	636	792	
	CDD50	924	0	1	1	12	64	136	258	278	128	36	8	2	
	CDD65	11	0	0	0	0	0	0	3	8	0	0	0	0	
	CDH74	74	0	0	0	0	6	3	23	40	2	0	0	0	
	CDH80	4	0	0	0	0	0	0	1	3	0	0	0	0	
Wind	WSAvg	7.7	8.4	9.6	8.8	8.0	7.6	7.6	7.7	7.3	7.0	7.0	6.9	7.1	
Precipitation	PrecAvg	38.3	4.5	3.4	3.1	2.3	2.5	2.5	2.4	2.9	2.9	3.9	3.7	4.3	
	PrecMax	53.5	9.3	9.1	8.1	5.1	5.3	5.5	4.6	6.2	8.3	7.4	7.3	9.9	
	PrecMin	25.0	0.7	1.0	0.9	0.4	0.7	0.8	0.9	0.7	0.8	0.9	0.9	0.7	
	PrecStd	6.1	2.1	2.0	1.7	1.1	1.1	1.2	1.2	1.3	1.5	1.6	1.5	2.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	53.6	56.0	62.9	73.9	71.9	77.3	78.8	70.4	63.4	57.7	54.9	
		MCWB	49.3	48.6	47.7	52.9	57.9	57.9	63.0	63.7	61.2	57.6	53.6	51.2	
	2%	DB	51.1	51.7	52.8	57.6	65.2	67.3	71.9	73.2	65.4	59.9	55.1	52.5	
		MCWB	47.9	48.4	47.6	49.8	54.6	57.1	61.7	63.0	58.0	54.7	51.9	49.5	
	5%	DB	49.4	50.1	51.0	55.0	60.9	63.9	67.2	69.1	62.7	57.5	53.5	50.4	
		MCWB	46.2	46.9	46.8	48.4	52.9	55.3	59.5	60.7	57.0	53.3	50.4	47.6	
	10%	DB	47.3	48.3	49.4	52.9	57.6	61.0	64.6	65.9	60.5	55.5	51.7	47.8	
		MCWB	44.3	44.9	45.7	47.5	51.0	53.8	58.4	59.3	55.5	51.8	48.6	45.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	50.3	50.1	53.5	59.1	60.3	64.9	66.3	62.8	58.7	54.6	52.5
			MCDB	52.2	52.4	52.8	59.9	70.0	67.7	73.4	75.2	68.4	60.6	56.7	54.2
2%		WB	48.6	48.8	48.8	51.2	55.9	58.0	62.5	63.6	59.9	55.5	52.6	49.7	
		MCDB	50.8	51.1	51.6	56.1	63.6	64.8	69.1	70.1	64.0	58.8	54.9	52.1	
5%		WB	46.3	47.2	47.6	49.6	53.6	56.6	60.7	61.9	57.9	53.8	50.7	47.6	
		MCDB	48.7	49.9	50.6	53.9	59.5	62.4	66.0	67.7	61.5	57.0	53.2	50.1	
10%		WB	44.7	45.2	46.0	48.0	51.9	55.1	59.1	60.1	56.0	52.1	48.9	45.4	
		MCDB	47.2	48.2	49.2	52.0	56.6	59.5	63.9	65.0	59.5	55.0	51.3	47.8	
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.4	9.2	10.7	11.9	11.1	11.2	11.7	10.7	8.8	7.2	7.0	
		MCDBR	8.4	8.2	11.0	15.1	18.3	16.5	16.8	17.3	14.1	11.4	8.8	10.1	
		MCWBR	7.7	6.8	7.3	9.1	9.2	8.6	8.9	8.7	8.2	8.1	7.5	8.6	
	5% WB	MCDBR	8.3	8.1	9.9	11.9	16.7	14.4	15.0	15.7	12.4	10.3	8.8	9.5	
		MCWBR	8.0	7.4	7.8	8.4	9.3	8.7	8.7	8.9	8.5	8.6	8.1	8.4	
Clear-Sky Solar Irradiance	taub	0.324	0.344	0.368	0.388	0.393	0.387	0.388	0.372	0.359	0.344	0.322	0.313		
	taud	2.170	2.243	2.242	2.242	2.284	2.341	2.350	2.401	2.430	2.380	2.248	2.127		
	Ebn at Noon	163	218	246	260	267	271	267	264	249	218	165	129		
	Edh at Noon	17	25	32	37	38	36	36	32	27	22	16	14		
All-Sky Solar Radiation	RadAvg	122	315	636	1068	1429	1391	1316	1046	729	384	163	88		
	RadStd	19	41	55	82	118	143	135	75	58	45	15	10		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page